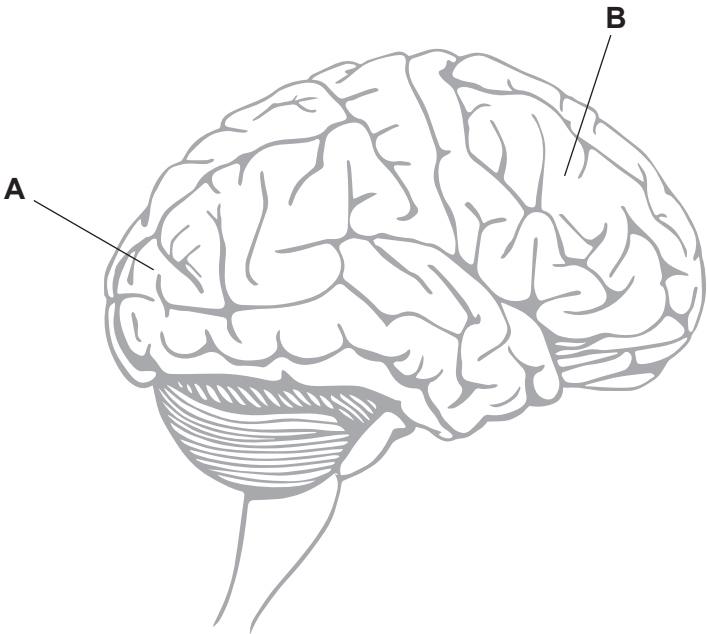


Option C: Neurobiology and Behaviour

8. (a) The image below shows the divisions of the human cerebral cortex into lobes.



(i) State the name and function of the lobes labelled **A** and **B** on the diagram above. [3]

	Name	Function
A		
B		

(ii) The brain can be studied in several ways. Outline the differences in the information provided by electroencephalography (EEG) and computerised tomography (CT). [2]

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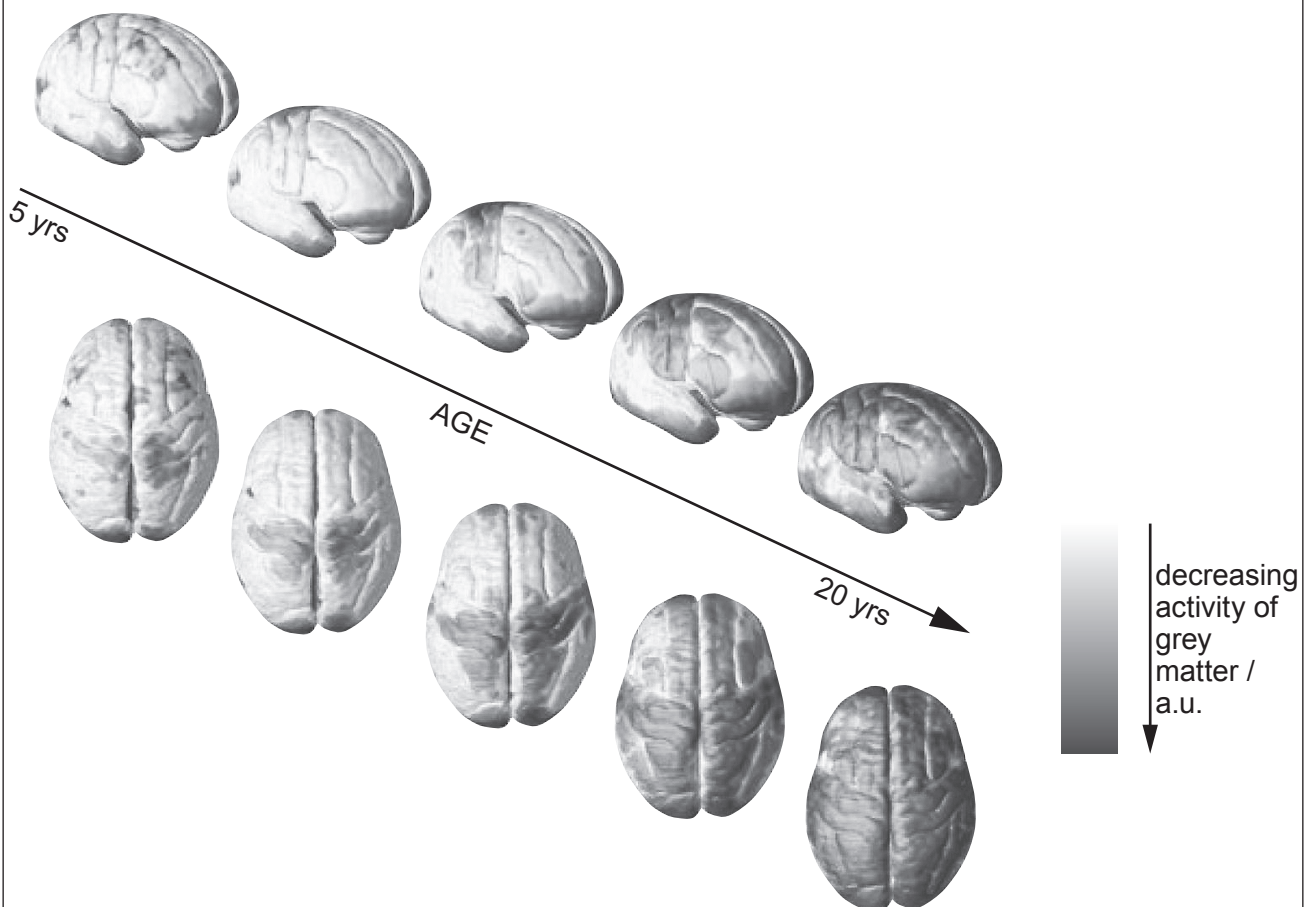
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- (b) Brain structure changes throughout life with most changes occurring in childhood. During childhood there are critical periods of development where synapses are strengthened in response to environmental stimuli. The critical period for the development of language is between 0-5 years.

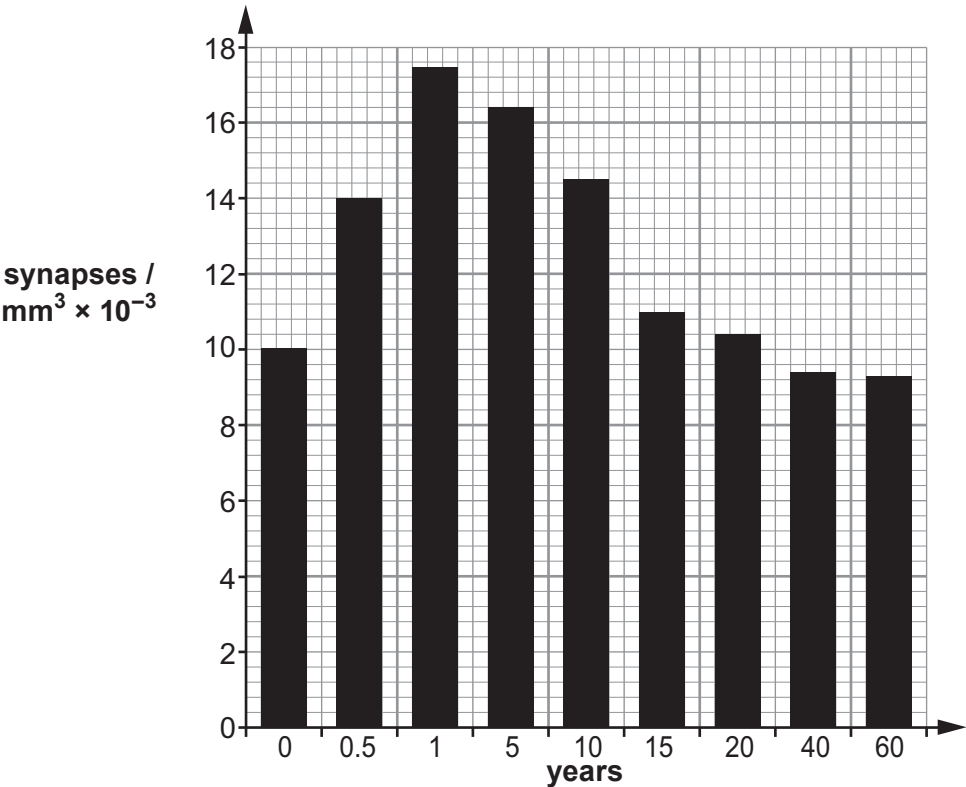
Exposure to neglect or psychological trauma during these critical periods in childhood, can change the normal developmental pattern e.g. in 'feral' children.

Brain activity in children was measured using functional magnetic resonance imaging (fMRI) to show the level of activity in the grey matter. The diagram below shows how the level of activity of grey matter in the brain changes with age after this critical period in a person who had not been exposed to neglect or psychological trauma.



Examiner
only

The graph shows the number of synapses present in the brain and the changes which normally occur as a child gets older.



- (i) Suggest why children who are not exposed to language during the critical period are likely to never develop speech. [4]

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- (ii) Suggest how the fMRI images and graph would differ for a feral child. [1]

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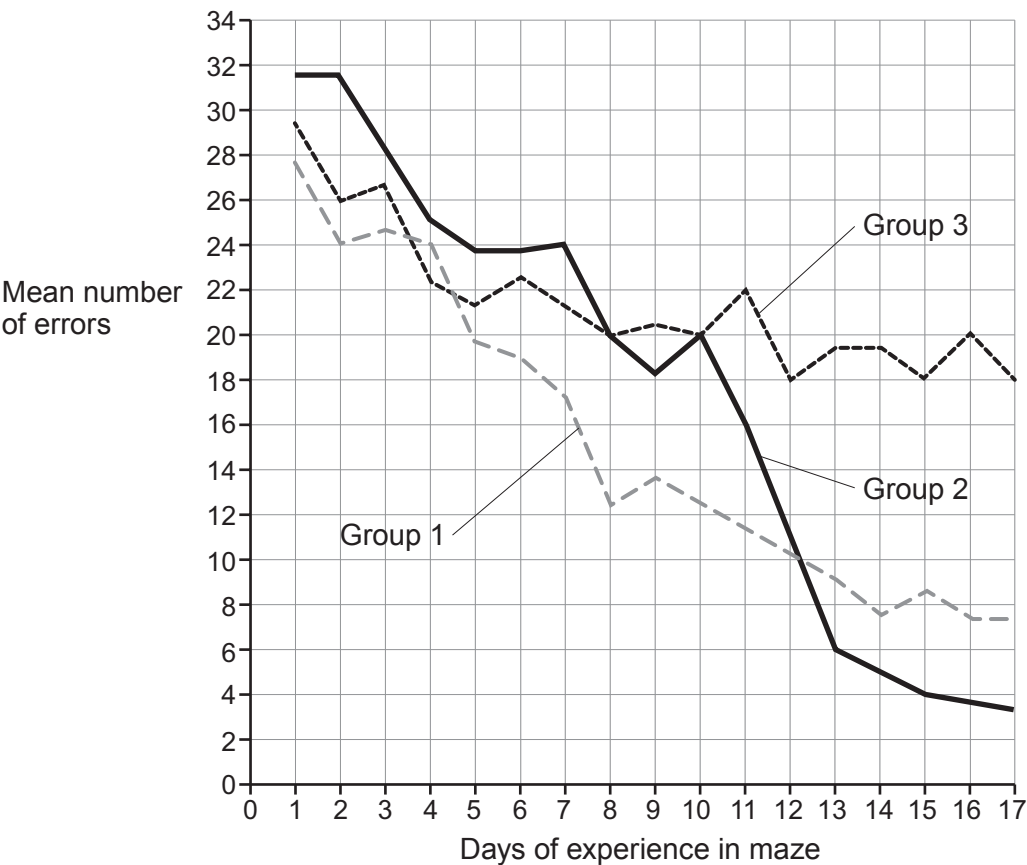
- (iii) State **one** factor which could be involved in causing this effect in feral children or those exposed to psychological trauma at an early age. [1]

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(c) In 1930, Tolman and Honzik investigated learning in rats. Thirty female rats were placed in three groups and the number of errors they made when going through the maze was recorded.

- Group 1 – Rewarded every time they completed the maze
- Group 2 – Placed in the maze every day and only rewarded from day 10 onwards
- Group 3 – Never rewarded



(i) Which area of the brain is involved in learning?

[1]

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Examiner
only

- (ii) Use the information provided to conclude the learning patterns exhibited by each group of rats. Explain your conclusions. [4]

Group 1

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Group 2

.....

.....

Group 3

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- (iii) Calculate the percentage change in the number of errors for group 2 between day 10 and day 13. [2]

Percentage change = %

- (iv) State **two** factors which should have been controlled in this investigation. [2]

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END OF PAPER

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